

Exercise 4 (16 points) – individual work

- The answers can be typed or handwritten (handwriting must be clear and readable), in this exercise sheet or your own sheet (put your name & ID at the top of the sheet). All answers must be saved to only 1 PDF file.
 - Some questions also require the submission of processes/workflows (file.rmp or file.ipynb).
 - In case of re-submission (after first grading) or submission after solution is given, your points will be weighted by 0.5.
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Two classifiers are used to classify 60 customer records whether they would buy sedan, pickup, or SUV. Their confusion matrices are

Classifier 1			
	Predicted Sedan	Predicted Pickup	Predicted SUV
Actual Sedan	15	4	5
Actual Pickup	4	14	2
Actual SUV	3	3	10

Classifier 2			
	Predicted Sedan	Predicted Pickup	Predicted SUV
Actual Sedan	18	3	3
Actual Pickup	3	10	7
Actual SUV	6	2	8

Answer all questions except (2.7) and (3.7) in 4 decimal places.

1. (Total 2 points) Consider overall performance.

1.1 Overall accuracy of classifier 1 = $(15+14+10)/60=0.6500$	1.2 Overall accuracy of classifier 2 = $(18+10+8)/60=0.6000$
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2. (Total 7 points) Consider performance of predicting class **Sedan**.

2.1 Precision of classifier 1 = $15/(15+4+3)=0.6818$	2.4 Precision of classifier 2 = $18/(18+3+6)=0.6667$
2.2 Recall of classifier 1 = $15/(15+4+5)=0.625$	2.5 Recall of classifier 2 = $18/(18+3+3)=0.7500$
2.3 F-measure of classifier 1 = $(2*0.625*0.6818)/(0.625+0.6818)=0.6522$	2.6 F-measure of classifier 2 = $(2*0.75*0.6667)/(0.75+0.6667)=0.7059$

2.7 By considering only precision, recall, and F-measure in (2.1)-(2.6), which classifier is better at predicting Sedan?

Classifier 1 has a slightly better overall accuracy and precision, but at the same time, classifier 2 has marginally better recall and F1 score. In this case, classifier 2 is better since higher f1 show that precision and recall is more balanced

3. (Total 7 points) Consider performance of predicting class **Pickup**

3.1 Sensitivity of classifier 1 = $14/(14+4+2) = 0.7000$	3.4 Sensitivity of classifier 2 = $10 / (10+3+7) = 0.5000$
3.2 Specificity of classifier 1 = $33 / (33+7) = 0.8250$	3.5 Specificity of classifier 2 = $35 / (35+5) = 0.8750$
3.3 Youden's J index of classifier 1 = $0.7 + 0.8250 - 1 = 0.5250$	3.6 Youden's J index of classifier 2 = $0.5+0.8750-1 = 0.3750$

3.7 By considering only sensitivity, specificity, and Youden's J index in (3.1)-(3.6), which classifier is better at predicting Pickup?

Classifier 1 has better sensitivity and Youde's index and the specificity of classifier 1 is also not that low compared to classifier 2. Both sensitivity and specificity of classifier 1 is good and Youden's also show that both of them are balanced between correct prediction and reject the wrong class.